

Session 4

Case Study: The United Nations and its technological and data challenges

Ernest Guevarra 

ernest@guevarra.io

University of Oxford



Costly and fragmented services, lagging in a digital age

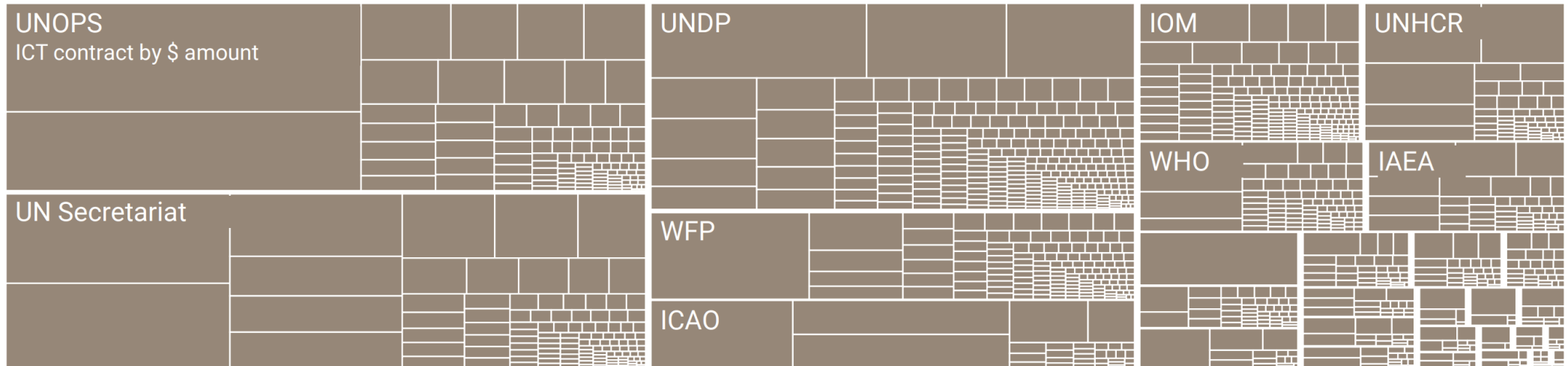
The problem

- The UN system's technology infrastructure is fragmented, expensive, and not strong enough for today's digital demands.
- Digital advancements, including artificial intelligence, are reshaping the world, but the UN's tech capabilities have not kept pace.
- Over **\$2 billion** is spent each year on software, cloud services, connectivity, and related support.

The problem (continued)

- Much of this spending happens in isolation, without scale, leading to higher costs, duplication, and weak interoperability.
- These inefficiencies slow the UN's ability to modernise and build future-ready capabilities.
- To remain credible and effective, the UN system must accelerate its digital adaptation.

Costly and fragmented technology



The solution

- The UN aims to harness technology responsibly to transform all aspects of its work.
- Shared ICT services and the UN 2.0 Agenda will guide this transformation.
- A new **Technology Accelerator Platform** will act as the system's engine for change.
- It will modernise internal UN operations.
- It will also enhance the UN's ability to support Member States in adopting digital and AI solutions effectively and responsibly.

Fragmented data, missed opportunities

The problem

- The UN system serves as a major custodian of global public data, statistics, and insights on topics ranging from population trends to climate risks.
- Despite this, its data platforms remain fragmented and duplicative, even as demand for faster and clearer information grows.
- Individual UN entities have built separate systems, resulting in parallel capacities that rarely connect.

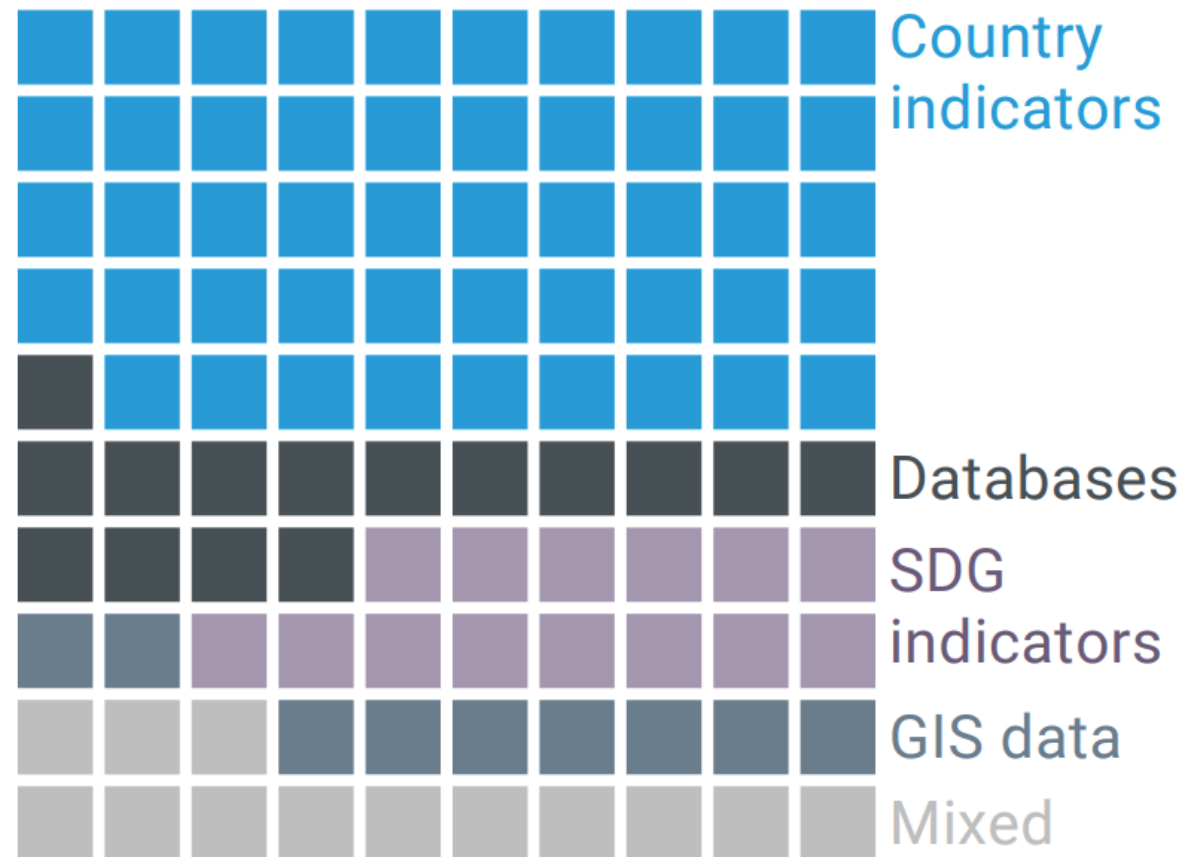
The problem (continued)

- Without interoperability, data cannot be shared effectively across entities, leading to blind spots, wasted effort, and missed opportunities to address interconnected global challenges.
- Data becomes far more useful when it is connected, but the capabilities needed to achieve this are limited, under-incentivised, and often hindered by entities' reluctance to collaborate.

Multiple public portals

~90

public data portals or sites



The solution

- New data initiatives will be developed across all UN pillars integrated into a **UN System Data Commons** built on shared data infrastructure.
- The shared backbone will be designed for full interoperability and able to generate insights at local, regional, and global levels.
- Entities will retain only the data that must remain separate, while all shareable data will be openly shared.

The solution (continued)

- To make this transformation sustainable, interoperability will be supported by collective financing and clear governance incentives.
- This approach will help connect resources and systems more effectively, increase impact, and reduce overall costs.